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# When Is Dyadic Transformation Reliable? A Threshold for Relational Event Models with Multi-Receiver Events

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## Abstract

The Relational Event Model (REM) is the standard framework for analysing relational event histories, but it is defined entirely on directed dyads, whereas many real communication processes are multi-receiver, meaning that a single sender addresses several receivers in one event. A common workaround is to transform each multi-receiver event into several dyadic events separated by a small tie-breaking timestamp noise, yet the proportion of multi-receiver events up to which this transformation still recovers the underlying parameters has not been established. We answer this with a controlled simulation in which the dyadic generating parameters are known. Baseline dyadic event histories ($N = 20$ actors, $M = 2{,}000$ events) are perturbed by randomly tagging an additional receiver onto a controlled proportion of events, the dyadic transformation is applied, and the refitted REM coefficients are compared against the known truth across 28 independent baseline datasets, with a native Relational Hyperevent Model reported as a contextual benchmark. Acceptability is judged by a pre-specified per-effect recovery-noise band. The result is a three-zone operating range: recovery is reliable up to a multi-receiver share of about 8%, borderline between 8% and roughly 12%, and unreliable above 12%, with the triadic outgoing-two-paths effect binding first. For shares above roughly 12% we recommend a native multi-receiver model such as the relational hyperevent model, a restructuring of the data, or another estimator designed for polyadic events, in place of the dyadic transformation.

**Keywords:** relational event model, relational hyperevent model, multicast, dyadic transformation, parameter recovery, simulation study

# Table of Contents

# 1. Introduction

Social interactions between actors are a foundational object of study in the social and behavioural sciences. The patterns by which actors initiate, return, and chain interactions over time are widely understood to encode the relational structures through which information, influence, and coordination move (Butts, 2008; Newman, 2018). Among the kinds of data that capture such interactions, relational event history (REH) data is distinctive in that each interaction is recorded as a discrete time-stamped event, preserving both the order and the precise timing of what happens (Butts, 2008; Butts & Marcum, 2017). The increasing availability of REH data, driven by email systems, instant messaging, sensor-based contact recording, and large-scale automated data collection, has made it possible to study social processes at fine temporal resolution (Lazer et al., 2009; Lerner & Lomi, 2020).

The Relational Event Model (REM) is the standard statistical framework for analysing REH data (Butts, 2008). It models the rate at which each potential interaction occurs as a log-linear function of statistics computed from the past history of events (endogenous statistics) and from actor or dyad attributes (exogenous statistics). The framework is dyadic by construction: every event is assumed to involve exactly one sender (the actor that initiates the interaction) and exactly one receiver (the actor the interaction is directed at), a pair referred to as a dyad (Butts, 2008). This assumption simplifies the model considerably, but it does not match the structure of many real interaction systems. Email and group meetings, for example, routinely involve a single sender addressing multiple receivers in one event. When such data is transformed into a dyadic format (typically by replicating each multi-receiver event as several single-receiver events), information about the multi-receiver structure of the original event is lost (Mulder & Hoff, 2024).

A native model for multi-receiver events is the Relational Hyperevent Model (RHEM), which extends REM to events involving any number of receivers (Lerner & Lomi, 2023). Despite this advance, dyadic transformation remains a common practical step, often applied without explicit justification (Mulder & Hoff, 2024). Lerner and Lomi (2023) report a better fit for RHEM than for a REM fitted to the dyadically transformed version of the same data. What has not been established is the size of the multi-receiver share up to which dyadic transformation still recovers the underlying REM parameters reliably. This is a clear gap: although the qualitative cost of dyadic transformation is documented, there is no controlled measurement of how much multi-receiver content the transformation can absorb before its estimates can no longer be trusted. Using a controlled simulation experiment with known dyadic generating parameters, this thesis estimates that share: the proportion of multi-receiver events up to which the dyadic transformation can still be relied upon for REM parameter recovery, and beyond which it cannot.

## 1.1 Social networks and relational event history data

A social network is a collection of social actors connected by some relation, where the actors can be individuals, groups, or organisations and the relation can be friendship, kinship, communication, or any other form of social tie (Wasserman & Faust, 1994; Newman, 2018). Social network analysis studies the patterns and structures of these relations, asking why ties form, how they persist, and how they change over time. A wide range of theoretical perspectives sit within this framework, from static structural analyses of fixed ties to fully dynamic accounts of evolving social processes (Snijders, 2001; Carrington et al., 2005).

REH data is one of the dynamic forms of social network data. It records a time-ordered sequence of directed interaction events between actors. Each event consists at minimum of three components: a sender, a receiver, and a time stamp at which the interaction took place (Butts & Marcum, 2017; Marcum & Butts, 2015). Senders and receivers can in principle be of any kind, and the action represented by an event can be any brief interaction. Multiple events ordered by time form a relational event history. Table 1 illustrates the structure on the first five rows of a synthetic example.

**Table 1**

*Illustrative Example of Relational Event History (REH) Data*

| Time | Sender | Receiver |
|---|---|---|
| 10.0 | 1 | 2 |
| 12.5 | 2 | 1 |
| 18.7 | 1 | 3 |
| 22.3 | 3 | 1 |
| 27.1 | 2 | 3 |

*Note.* The first five rows of a synthetic REH dataset. Time is given in arbitrary continuous units. Sender and Receiver columns identify actors by integer code.

REH data differs from snapshot or panel network data in two important ways. First, the unit of observation is a single discrete interaction rather than a state or an aggregate count. Second, exact timing is preserved, which allows process-oriented questions about what drives the next event in a sequence to be addressed directly (Butts, 2008; Lakdawala et al., 2025). These properties make REH data well-suited to studying mechanisms such as inertia, reciprocity, and turn-taking effects in conversation (Butts, 2008; Gibson, 2003). They also make REH data hard to analyse with conventional regression methods, which do not naturally accommodate the temporal sequencing and inter-event dependence of relational events (Butts, 2008; Butts & Marcum, 2017).

## 1.2 The relational event model

REM, introduced by Butts (2008), is the most widely used framework for analysing REH data; for a recent comprehensive survey, see Bianchi et al. (2024). REM treats each potential directed dyad as competing to fire next: at any moment in continuous time, every dyad has an instantaneous rate (a hazard) of producing the next event, and these competing rates are what the model parameterises. The rate for each dyad is a log-linear function of statistics computed from past events and from actor or dyad covariates; the formal hazard expression, together with the baseline pacing constant $\beta_0$, is given with the data-generating process in Section 2.1. The log-linear form is a standard hazard-model choice: the exponential keeps the rate strictly positive for any combination of statistic values, and each statistic contributes multiplicatively to the rate, so its parameter has a direct effect-size reading. The risk set at time $t$ is the collection of all dyads that could fire at that moment; for a closed network of $N$ actors, it contains $N(N-1)$ directed dyads. Between events the rates stay fixed; when an event fires, the statistics are updated to reflect the new history and the rates are recomputed.

The statistics in the rate function fall into two classes. Endogenous statistics are computed from the history of past events. Examples include inertia, the standardised count of past events from $i$ to $j$, which captures the tendency to repeat past interactions; reciprocity, the count of past events from $j$ to $i$, which captures the tendency to return them; and triadic closure effects such as outgoing two-paths or transitive closure (Butts, 2008). Exogenous statistics encode actor or dyad attributes that do not change with the event history, such as gender, role, department, or pairwise homophily on a categorical variable.

Two likelihood forms are standard (Butts, 2008): the full likelihood, which uses inter-event durations explicitly and identifies the absolute rate scale (and therefore $\beta_0$), and the ordinal likelihood, which conditions only on event order and discards the timing information that is available when exact timestamps are known (Meijerink-Bosman, Back, et al., 2022). This thesis uses the full likelihood throughout. Because the full likelihood reads information from the exact time elapsed between consecutive events, it requires that no two events share a timestamp; the multi-receiver transformation introduced in this thesis is designed so that this requirement continues to hold (Section 2.3).

## 1.3 The multi-receiver challenge and the Relational Hyperevent Model

REM is defined entirely on dyads: the risk set, the rate function, and the entire family of endogenous statistics are all defined on directed pairs $(i, j)$ (Butts, 2008). Many real interaction systems do not satisfy this assumption. Lerner and Lomi (2023) refer to interaction processes that involve more than two actors per event as polyadic, and document them across a wide range of social-science settings, from scientific citation networks to legal precedent chains and group-mediated disease transmission. Mulder and Hoff (2024) report that 31% of all messages in the Enron email dataset are multicast (a single sender addressing multiple receivers) with $M = 21{,}635$ messages sent among $N = 156$ actors, and explicitly characterise this 31% as a relatively low multicast share for organisational email, lower than in many comparable communication datasets. Multicast is therefore common enough in real communication data that how it is handled is a routine and consequential modelling decision for applied REM analysis.

When such data is fitted with a dyadic REM, each multi-receiver event is first replicated as several dyadic events, one per receiver; Mulder and Hoff (2024) document this replication as the standard workaround in applied work and identify it as the proximate source of the information losses critiqued in their paper. The replication duplicates the receiver-set composition into an unordered collection of pairs, and because the duplicated dyads share the original event time, it creates simultaneous events that violate the exact-timing likelihood. The most common practice keeps these tied timestamps and works with event order alone; the variant adopted in this thesis instead adds small random perturbations so that the events fire sequentially, which preserves identifiability of the pacing constant required by the full likelihood of Butts (2008) at the cost of introducing an arbitrary ordering among the co-receivers of each multi-receiver event. Table 2 illustrates this transformation on a single event.

**Table 2**

*Illustrative Multi-Receiver Event and Its Dyadic Transformation*

| Stage | Time | Sender | Receiver(s) |
|---|---:|:---:|:---:|
| Original multi-receiver event | 18.7 | 1 | 2, 3 |
| Dyadic transform, first row | 18.7000 | 1 | 2 |
| Dyadic transform, second row | 18.7004 | 1 | 3 |

*Note.* A single event in which actor 1 addresses actors 2 and 3 at time 18.7 (top row) is replicated into two single-receiver events. A small tie-breaking timestamp noise (here 0.0004) separates the duplicated timestamps so that the two events fire sequentially, which the full-likelihood REM requires (Section 2.3). After the transformation the original receiver set $\{2,3\}$ appears only as two separate dyads, so the fact that actors 2 and 3 were addressed together in one message is no longer recoverable.

Mulder and Hoff (2024) identify three concrete information losses from this replication. First, the receiver-set composition is replaced by an unordered collection of dyads, so it is no longer possible to tell which receivers were addressed together in the same message, and any distinction in how each receiver figured in that message is lost. Second, receiver-set cardinality is removed, even though the number of receivers per event is itself an informative feature of any multicast process. Third, the apparent sample size is inflated, which can lead to overconfident inferences with under-estimated standard errors.

The natural alternative within the same hazard framework is the Relational Hyperevent Model (RHEM) of Lerner and Lomi (2023), which generalises REM to directed hyperedges: events that connect a sender to a receiver set of arbitrary size. The event rate is then a function of covariates defined on the entire receiver set, not on a sum over dyads. Crucially, this allows RHEM to express patterns that have no dyadic analogue, the clearest being group-level co-participation history, the extent to which a particular subgroup of receivers has previously been addressed jointly by the same sender. The open-source `eventnet` software is the standard implementation (Lerner & Lomi, 2023).

RHEM serves as a contextual benchmark in this thesis: we fit it to the same multi-receiver synthetic data as the transformed REM and report its estimates alongside. The threshold itself is decided against the known dyadic generating parameters, not against the RHEM estimates.

## 1.4 Methodological grounding

Three further bodies of work, although not specific to multi-receiver REM, ground the methodological decisions made in this thesis.

The first is statistic comparability across REM and RHEM. A precondition for a meaningful comparison between transformed REM estimates and a RHEM benchmark is that the statistics are defined as closely as possible in both models; otherwise observed differences cannot be attributed to the transformation alone. A cross-mapping between the dyadic `remstats` catalogue (Meijerink-Bosman et al., 2023) and the corresponding RHEM effects documented in the `eventnet` reference guide (Lerner & Lomi, 2023) shows that inertia, reciprocity, dyadic attribute difference, and same-group homophily map closely between the two frameworks, although they are not in every case strictly identical; transitive closure via outgoing two-paths corresponds only up to a counting-convention offset. These five effects serve as the primary endpoints in this thesis.

The second is simulation-based validation. Across the REM literature, the standard methodological approach is to generate synthetic event histories from known parameters, fit the model to the synthetic data, and evaluate parameter recovery under controlled conditions (Meijerink-Bosman, Leenders, et al., 2022; Mulder & Leenders, 2019). Lakdawala et al. (2025) give an explicit methodological treatment of REH simulation, and their accompanying R package `remulate` is the simulation engine used in this thesis.

The third is principled perturbation of temporal-network data. Recent work in temporal networks (Gauvin et al., 2018; Longa et al., 2024) recommends a simple rule when modifying event-stream data: state up front which features of the data should be kept fixed, and randomise only the rest.

## 1.5 Current research

REM is a well-established framework for dyadic interaction sequences, with mature applied workflows and dynamic extensions that allow for time-varying parameters. Its dyadic assumption, however, is incompatible with the many communication settings in which a single sender addresses several receivers in one event. RHEM (Lerner & Lomi, 2023) provides a native polyadic alternative within the hazard framework, and Mulder and Hoff (2024) set out the information losses that dyadic separation incurs. Even so, the dyadic REM remains the default in applied work: it has mature and well-documented software (Meijerink-Bosman et al., 2023; Arena et al., 2023, 2024), a large body of methodological guidance (Butts, 2008; Meijerink-Bosman, Back, et al., 2022; Bianchi et al., 2024), and a familiar interpretation, and the dyadic transformation lets applied researchers reuse that entire toolchain on multi-receiver data without committing to a less established model. What the field still lacks is a controlled, quantitative answer to the question this convenience raises: the proportion of multi-receiver events up to which dyadic transformation can still be relied upon for REM parameter recovery, and beyond which it cannot.

This thesis aims to provide an answer. The primary research question is:

> Up to what proportion of multi-receiver events does separating each multi-receiver event into dyadic events with a small tie-breaking timestamp noise preserve REM parameter estimates that remain close to the known dyadic generating parameters?

The study uses a simulation-first design with known data-generating parameters (the generating model and its parameter values are given in Section 2.1), and evaluates the recovered REM estimates against those parameters across a controlled range of multi-receiver proportions. The threshold rule, the per-effect band against which it is applied, and the cross-dataset averaging that smooths the estimate across independent baseline generations are specified in the Methodology chapter.

The threshold is the contribution this thesis offers to researchers and analysts working with communication data that includes multi-receiver events: the multi-receiver proportion below which dyadic REM with a tie-breaking timestamp noise still recovers parameters close to the truth, and above which it can no longer be relied upon. The simulation is designed to identify where the dyadic transformation fails; what to do once that point is passed is outside the scope of the experiment and is taken up in the Discussion.

The remainder of the thesis develops the study as follows. The Methodology chapter describes the synthetic-generation procedure, the multi-receiver construction, the REM and RHEM estimation pipelines, and the acceptability rule under which the threshold is determined. The Results chapter reports the threshold finding and its sensitivity to the baseline generation. Finally, the Conclusion and Discussion chapter considers what this finding implies for applied research with communication data, the limitations under which the recommendation holds, and the ethical considerations attached to applying it.

# 2. Methodology

This chapter describes the synthetic data used in the thesis and the methods applied to it: the baseline data-generating process, the multi-receiver construction, the tie-breaking transformation back to dyadic form, the dyadic REM and RHEM estimation pipelines, the cross-dataset replication, and the acceptability rule under which the threshold is determined.

## 2.1 Data source and baseline data-generating process

This thesis uses synthetic relational event history (REH) data exclusively; no empirical dataset is analysed. The unit of analysis is a baseline dataset: a single dyadic REH, that is, one event sequence in which every event has exactly one receiver, that serves as the unaltered starting point before any multi-receiver events are added. The study generates many such baseline datasets, one per independent simulation seed (Section 2.6). Each is simulated with the `remulate` R package (Lakdawala et al., 2025), which implements the dyadic-REM simulation framework of Butts (2008), generating events sequentially under the REM hazard model with each new event conditioned on the cumulative history of past events.

Under this model, every directed dyad competes to produce the next event, and the instantaneous rate at which sender $i$ addresses receiver $j$ at time $t$ is a log-linear function of the model statistics,

$$\lambda(i,j,t) = \exp\left(\beta_0 + \sum_{q=1}^{Q} \beta_q X_q(i,j,t)\right),$$

where the index $q = 1, \dots, Q$ runs over the statistics included in the model, $X_q(i,j,t)$ is the value of the $q$-th statistic for the dyad $(i,j)$ at time $t$, $\beta_q$ is its corresponding parameter, and $\beta_0$ is the baseline pacing constant (Butts, 2008). Between events the rates stay fixed; when an event fires, the statistics are updated to reflect the new history and the rates are recomputed.

The configuration used throughout the thesis fixes the actor pool at $N = 20$ and the event count at $M = 2{,}000$, giving $N(N-1) = 380$ directed dyads in the risk set at every moment. The baseline draws two actor-level attributes that the model uses as covariates: a binary group indicator (sampled independently for each actor with equal probability), and a continuous activity score (sampled independently from a standard normal distribution). The statistics that enter the rate function and the true parameter values $\beta_q$ from which every baseline is generated are listed in Table 3. Because these values are set by us before any data is simulated, they are the known generating truth against which parameter recovery is later judged.

**Table 3**

*Effects and True Parameter Values Used in the Dyadic Baseline Simulation*

| **Effect** | **Type** | **Description** | **True $\boldsymbol{\beta}$** |
|---|---|---|---|
| Baseline | Pacing constant | Log of the per-dyad baseline rate | $-5.00$ |
| Inertia | Endogenous, dyad-level | Standardised count of past $i \to j$ events | $0.02$ |
| Reciprocity | Endogenous, dyad-level | Standardised count of past $j \to i$ events | $0.10$ |
| OTP | Endogenous, triadic | Outgoing two-paths $i \to h \to j$ (transitive closure) | $0.03$ |

| | | | |
|---|---|---|---|
| Difference (active) | Exogenous, dyad-level | Absolute difference $\|a_i - a_j\|$ on the continuous attribute | $0.15$ |
| Same (group) | Exogenous, dyad-level | Indicator of shared group membership | $0.40$ |

*Note*. The baseline parameter is on the log-rate scale, so $\exp(-5) \approx 0.0067$ is the per-dyad rate when every other statistic is zero. The inertia value of $0.02$ is calibrated to the $(N = 20, M = 2{,}000)$ configuration: cumulative-count inertia is positive feedback, so larger values compound across the 2,000-event sequence until expected waiting times collapse and remaining events fire at duplicate timestamps. The feasible region for $\beta_{\text{inertia}}$ is therefore $M$-dependent.

The simulator is configured to produce exactly $M = 2{,}000$ events with no random warm-up. Each baseline is checked for duplicate timestamps before it enters the analysis.

## 2.2 Multi-receiver construction

Multi-receiver versions of the baseline are produced by additive random tagging. The procedure starts from the original dyadic baseline (every event has one receiver), tags a target proportion $p$ of events sampled uniformly at random without replacement, and adds one extra receiver to each tagged event. This proportion of multi-receiver events, $p$, is the study's multi-receiver (MR) proportion; each condition is indexed by its MR level (its value of $p$ in per cent). The extra receiver is drawn uniformly at random from the actors who are neither the sender nor the original receiver of that event. No baseline event is removed, repositioned, or otherwise altered; the construction is purely additive. We avoid holding the long-table row count fixed by deletion because deletion would over-represent high-activity dyads in the retained sample, inflating the cumulative-count statistics in a way that would be indistinguishable from a genuine multi-receiver effect.

Five conditions are evaluated. The 0% condition is the unaltered baseline. Four multi-receiver conditions are constructed at $p \in \{0.02, 0.08, 0.16, 0.24\}$. Together with the 0% reference, these five MR levels, $\{0,2,8,16,24\}\%$, form the grid on which the analysis is run. The upper end sits below the 31% multicast share that Mulder and Hoff (2024) report for the Enron corpus, so the grid covers the region where an analyst is most likely to expect the dyadic transformation to hold. The recommendation derived from the grid is correspondingly bounded above by $p = 0.24$, with no claim made about higher proportions. Table 4 summarises the resulting conditions.

**Table 4**

*Multi-Receiver Conditions Produced From One Baseline*

| Condition | Target $p$ | Baseline events ($M$) | Tagged events | Long-table rows |
|---:|---:|---:|---:|---:|
| 0% | 0.00 | 2,000 | 0 | 2,000 |
| 2% | 0.02 | 2,000 | 40 | 2,040 |
| 8% | 0.08 | 2,000 | 160 | 2,160 |
| 16% | 0.16 | 2,000 | 320 | 2,320 |
| 24% | 0.24 | 2,000 | 480 | 2,480 |

*Note*. Long-table rows are computed as baseline events plus tagged events (one extra-receiver row per tagged event). The target proportions $\{0.02, 0.08, 0.16, 0.24\}$ were chosen so that $p \cdot M$ is an integer at every level; the realised proportion therefore equals the target exactly. The 0% condition is identical to the baseline and serves as a parameter-recovery reference.

Two design choices shape the construction. First, every multi-receiver event has exactly two receivers; receiver-set cardinality is not varied. Second, the extra receiver is drawn uniformly at random from non-participating actors, so it carries no signal: it is uncorrelated with the sender's past interaction history and with any covariate. Both choices isolate the effect of multi-receiver injection itself from the effects of structured multicast that Mulder and Hoff (2024) identify as the main sources of information loss in real multicast data: for example, the same group of receivers being addressed together repeatedly, or receiver sets that are mostly small with an occasional very large one. The threshold reported in the Results chapter is therefore conditional on this random-tagging design; whether structured multicast would shift the threshold upward or downward is left to the Discussion.

## 2.3 Tie-breaking timestamp-noise transformation

A multi-receiver event with a sender $i$ and a receiver set $J = \{j_1, j_2\}$ is represented in the multi-receiver long table as $|J| = 2$ rows sharing the original event time $t$. The full likelihood of the dyadic REM requires that no two events share a timestamp (Butts, 2008); leaving the shared time in place would force the ordinal likelihood, which discards the inter-event-duration information that the dyadic generating process actually used. We therefore apply a tie-breaking timestamp-noise transformation that perturbs the duplicated timestamps just enough to break the tie while preserving the original event order.

The transformation treats every multi-receiver row as a candidate for noise injection. For each tagged event, the rows after the first are shifted by an independent draw from a uniform distribution on $(0, \varepsilon_{\max})$, where

$$\varepsilon_{\max} = \varepsilon_{\text{factor}} \cdot \min\{t_{(k+1)} - t_{(k)} : t_{(k)} \neq t_{(k+1)}\}.$$

The right-hand side is the minimum inter-event gap of the baseline among distinct timestamps, scaled by a fixed factor $\varepsilon_{\text{factor}} = 0.10$. This convention makes $\varepsilon_{\max}$ strictly smaller than the smallest gap between any two distinct baseline events, so the perturbed events of any tagged group remain inside their original inter-event interval and the relative order of all distinct-time events is preserved. Concretely, at $(N = 20, M = 2{,}000)$ the realised inter-event gaps are typically on the order of $10^{-1}$ to $10^{-2}$ time units, which puts $\varepsilon_{\max}$ on the order of $10^{-3}$: one to two orders below typical inter-event gaps, and an order of magnitude below the smallest distinct gap.

Because the perturbation is far smaller than the smallest distinct inter-event gap, it breaks the ties within each multi-receiver event without reordering any distinct-time events, leaving the long table compatible with the full likelihood of Butts (2008) and its absolute rate scale.

## 2.4 Dyadic REM estimation

The dyadic REM is fitted to the noise-perturbed long table from Section 2.3 using the standard R toolchain. The package `remify` (Arena et al., 2023) prepares the REH object from the perturbed long table; the package `remstats` (Meijerink-Bosman et al., 2023) then computes the five tie-oriented statistics specified in Table 3; and the package `remstimate` (Arena et al., 2024) then fits the model by maximum likelihood under the full (exact-timing) likelihood of Butts (2008). The risk set is the full $N(N-1) = 380$ directed dyads; no case-control sampling is applied on the REM side because at $M = 2{,}000$ events the full likelihood is tractable.

The recovered coefficient vector $\widehat{\boldsymbol{\beta}}^{\text{REM}}$ contains six estimated coefficients, one for each of the six generating coefficients of Table 3: the log baseline rate and the five tie-oriented effects. The acceptability rule of Section 2.7 operates on the five tie-oriented effects; the log baseline rate is recovered for completeness but is excluded from the rule for the reason given in Section 2.7.1. The bias of the REM estimate at multi-receiver proportion $p$ on a single dataset is the per-coefficient difference $\widehat{\boldsymbol{\beta}}_p^{\text{REM}} - \boldsymbol{\beta}$. We take the across-seed mean of this difference at each $p$ as the bias of interest for the cross-dataset analysis of Section 2.6.

## 2.5 RHEM benchmark

For each MR level, including the 0% condition, we additionally fit the Relational Hyperevent Model of Lerner and Lomi (2023) as a contextual benchmark. We use the standard `eventnet + survival` pipeline of Lerner and Lomi (2023): the `eventnet` software enumerates the hyperedge risk set and produces, for the observed hyperevents and a case-control sample of non-events (hyperedges that could have occurred but did not), the hyperedge covariates that correspond to the five dyadic statistics fitted on the REM side; the model is then fitted by stratified Cox regression in the `survival` package (Therneau, 2023). The estimates obtained this way are on the same RHEM coefficient scale used by Lerner and Lomi (2020, 2023). Stratification convention and pipeline details are given in Appendix A. The `eventnet` runs are configured per (seed × MR level) cell.

One caveat applies to the outgoing-two-paths effect. The dyadic and hyperedge pipelines count outgoing two-paths in slightly different ways, so the RHEM estimate of this single coefficient carries a small fixed offset relative to the REM coefficient scale. This offset is the same at every multi-receiver level, which is what marks it as a counting-convention difference rather than an effect of the transformation; its magnitude is characterised empirically in the Results chapter. Because the threshold is judged against the known generating parameters on the REM side (Section 2.7), this offset does not affect it.

## 2.6 Cross-dataset replication

A single baseline simulation is enough to demonstrate the threshold mechanism, but the resulting threshold estimate depends on characteristics of the particular baseline that was generated. To smooth out single-dataset variance, we run the analysis over 30 independent baseline simulations, indexed by consecutive integer seeds starting at `20260423`. For each seed a baseline is generated, the five multi-receiver conditions are constructed by additive random tagging (Section 2.2), the tie-breaking transformation of Section 2.3 is applied to each condition, and both the dyadic REM (Section 2.4) and the RHEM benchmark (Section 2.5) are fitted, writing the recovered coefficients to two long-format result tables, for REM and RHEM respectively, keyed by seed and MR level.

A baseline-quality guard inspects each baseline before any transformation or estimation is performed and skips any seed whose minimum inter-event gap is small enough to risk duplicate timestamps after the tie-breaking noise is added. Two of the 30 seeds fail this guard and are dropped before any analysis; the reported result is based on $n = 28$ clean independent datasets. Because the guard only inspects the baseline before any transformation, dropping seeds cannot bias the REM-versus-RHEM comparison downstream.

The dataset-averaged statistics are computed by simple unweighted arithmetic mean over the 28 retained seeds; the Monte Carlo standard error on the mean of any per-effect bias is the across-seed sample standard deviation divided by $\sqrt{28}$, and is used to attach Monte Carlo error bars to the headline plot. The full pipeline is deterministic in the baseline seed: every number reported in the Results chapter can be regenerated from the seed list alone.

## 2.7 Acceptability rule and threshold determination

The headline finding of this thesis is a threshold on the multi-receiver proportion: the largest $\mathrm{MR}$ proportion at which the dataset-averaged REM estimate of each of the five tie-oriented coefficients remains within a per-effect practical-significance band around its known generating value.

### *2.7.1 Per-effect recovery-noise band*

For each statistic, the band is defined as

$$\text{band}_{\text{stat}} = k \cdot \mathrm{SD}_{\text{stat}}^{(\mathrm{MR}=0)},$$

where $\mathrm{SD}_{\text{stat}}^{(\mathrm{MR}=0)}$ is the across-seed sample standard deviation of the recovered REM coefficient at the 0% multi-receiver condition, computed over the 28 retained baseline seeds described in Section 2.6. The multiplier $k = 2$ is pre-specified: it was fixed in advance of the threshold determination on 2026-06-11 and was not modified after the recovered numbers were inspected. The choice of $k = 2$ corresponds to approximately a 95% interval under near-normal recovery, the standard practical-significance convention.

$\mathrm{SD}_{\text{stat}}^{(\mathrm{MR}=0)}$ measures the parameter-recovery noise of the dyadic-REM simulator itself, the typical magnitude by which a single recovered REM coefficient deviates from the truth when nothing has been done to the data. The rule therefore asks whether the systematic bias introduced by the multi-receiver injection and the subsequent transformation is large relative to the recovery noise a single-dataset analyst would actually face. This is why the band is calibrated to single-dataset recovery dispersion rather than to the Monte Carlo standard error of the 28-seed mean, which shrinks with $\sqrt{n}$ and would instead measure the bias against averaging noise.

The same 28-seed pool is used both for the dataset-averaged bias estimate at each MR level and for the band derivation at $\mathrm{MR} = 0$ that anchors the acceptability rule. Because the 0% condition is identical to the baseline (Section 2.2, Table 4) and the band is only applied to the four transformed conditions, the rule does not use the same observations both to estimate the band and to evaluate against it.

The rule auto-scales with the magnitude of each parameter, which avoids the problem with a single fixed cap: a cap loose enough for the largest generating coefficient (same-group homophily, $0.40$) would be wider than the smallest (inertia, $0.02$). The five per-effect band values are reported in the Results chapter alongside the biases they are compared with. The bands are estimated quantities: at $n = 28$ each per-effect SD has roughly $\pm 14\%$ relative uncertainty, which propagates into the bias-to-band ratios and limits the precision to which the threshold can be quoted.

The log baseline rate $\beta_0$ is recovered alongside the five tie-oriented effects but is excluded from the rule. Because the additive construction of Section 2.2 inflates the long-table row count from $M$ to $M(1 + p)$, the recovered $\beta_0$ tracks the pacing of the augmented process and shifts by approximately $\log(1 + p)$; it therefore measures event-count inflation rather than a recovery failure, and answers a different question from the one the threshold poses. The derivation, its numerical predictions, and the conditions under which the shift is exact are given in Appendix B.

### *2.7.2 The rule-defining grid*

The per-effect recovery-noise band of Section 2.7.1 is applied on the pre-specified grid of multi-receiver proportions (Table 4). The headline threshold is the largest proportion on this grid at which every rule-governed REM coefficient stays inside its per-effect band.

Additional cells outside the rule-defining grid may be added to locate the threshold more precisely between adjacent grid points. These cells appear in the Results plots and tables but do not enter the threshold calculation. The cross-dataset workflow (Section 2.6) allows them to be added without rerunning the full grid, which we use to place a single $\mathrm{MR} = 12\%$ cell in the 8%–16% gap.

#### *2.7.3 Sensitivity to the choice of acceptability rule*

Two acceptability specifications were considered during the design of this thesis. The per-effect recovery-noise band is the primary specification under which the headline threshold is reported. An earlier specification, a single absolute log-rate cap of $|\text{bias}| \leq 0.05$ across all statistics, is reported alongside in the Results chapter as a one-line sensitivity check. The swap was motivated by the parameter-magnitude problem described above, not by any property of the recovered numbers.

# 3. Results and analysis

This chapter reports the threshold finding and the quantities that support it, beginning with the recovery-noise bands that set the scale for every later comparison and ending with a check that the threshold survives a change of acceptability rule. All averages are over the $n = 28$ retained baseline seeds unless stated otherwise.

## 3.1 Per-effect recovery-noise bands

Table 5 reports the per-effect band, $\text{band}_{\text{stat}} = 2 \cdot \mathrm{SD}_{\text{stat}}^{(\mathrm{MR}=0)}$, together with the recovery at the 0% condition that defines it. At baseline the dyadic REM recovers every generating coefficient well within its band; inertia and outgoing two-paths sit closest to their edges (at $0.21$ and $0.24$ of the band), the other three within $0.10$. The bands themselves span more than an order of magnitude, from $0.0031$ for outgoing two-paths to $0.0798$ for same-group homophily, which is the spread the per-effect calibration is designed to absorb: a single absolute cap wide enough for same-group homophily would be more than twenty times the recovery dispersion of outgoing two-paths.

**Table 5**

*Per-Effect Recovery-Noise Bands and Baseline Recovery (0% Condition, $n = 28$)*

| Effect | True $\boldsymbol{\beta}$ | Mean estimate | $\mathbf{SD}^{(\mathbf{MR=0})}$ | Band (2 SD) | \|bias\| / band |
|---|---|---|---|---|---|
| Inertia | 0.02 | 0.0171 | 0.0068 | 0.0135 | 0.21 |
| Reciprocity | 0.10 | 0.1014 | 0.0088 | 0.0175 | 0.08 |
| OTP | 0.03 | 0.0307 | 0.0015 | 0.0031 | 0.24 |
| Difference (active) | 0.15 | 0.1487 | 0.0318 | 0.0636 | 0.02 |
| Same (group) | 0.40 | 0.3927 | 0.0399 | 0.0798 | 0.09 |

*Note.* The band is twice the across-seed sample standard deviation of the recovered REM coefficient at the 0% condition (Section 2.7.1). The final column is the absolute mean bias at the 0% condition expressed as a fraction of the band; values below 1 are inside the band.

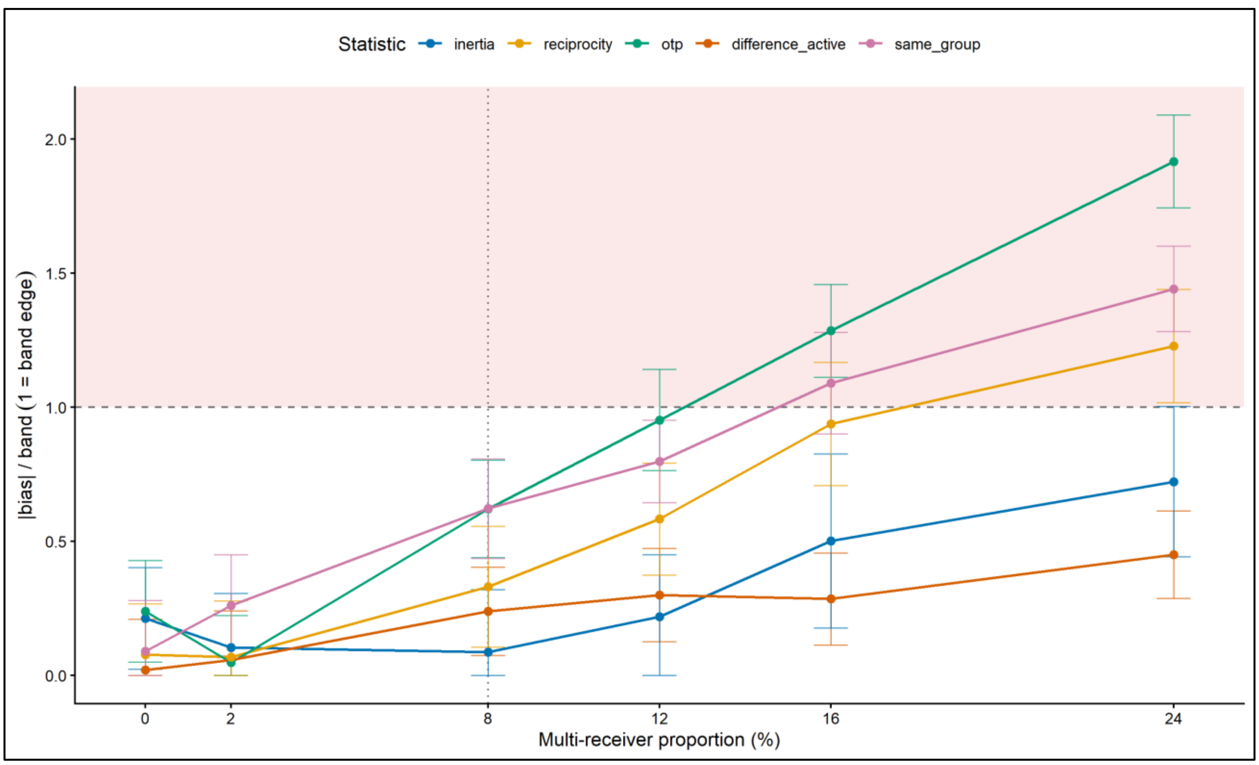
## 3.2 The headline threshold

Figure 1 shows the normalised bias-to-band ratio for each of the five tie-oriented effects across the multi-receiver grid; a ratio of 1 is the band edge, and anything above it is out of band.

The headline threshold is defined as the largest multi-receiver proportion on the rule-defining grid at which all five tie-oriented effects stay within their per-effect bands (Section 2.7). On the grid $\{0,2,8,16,24\}\%$, every effect stays within its band through the 8% condition, while at least one effect (outgoing two-paths) breaches its band at 16%. The headline threshold is therefore 8%: dyadic transformation with a tie-breaking timestamp noise recovers the generating parameters up to a multi-receiver proportion of 8%, and is no longer reliable by 16%.

**Figure 1**

*Normalised Threshold View: Bias-to-Band Ratio by Multi-Receiver Proportion ($n$ = 28 Seeds)*




*Note.* Points are dataset-averaged absolute biases divided by the per-effect band of Table 5, plotted for each of the five tie-oriented effects. The dashed horizontal line is the band edge; the shaded region above it is out of band; the dotted vertical line marks the 8% threshold.

Error bars are $\pm 2$ Monte Carlo standard errors divided by the band. The 12% condition is an off-grid descriptive cell (Section 2.7.2) and does not enter the threshold calculation.

## 3.3 Per-effect bias trajectories and the 12% boundary

Table 6 gives the bias-to-band ratio for every effect at each multi-receiver proportion. Two effects govern the threshold. Outgoing two-paths and same-group homophily rise fastest with the multi-receiver proportion and are the first to leave their bands, at the 16% condition (1.29 and 1.09 respectively); reciprocity follows by the 24% condition (1.23). Inertia and the active-attribute difference never breach their bands on the grid, reaching at most 0.72 and 0.45 at 24%. The direction of the bias is consistent: difference, reciprocity, outgoing two-paths, and same-group homophily all attenuate towards zero as multi-receiver events are added, whereas inertia is recovered marginally low at baseline (0.21 of its band) and drifts upward to a mild inflation by the upper conditions.

The off-grid descriptive cell at 12% locates the deterioration more precisely. At 12% all five effects are still within band, but outgoing two-paths sits at 0.95 of its band, against 0.62 at 8%. The transition from reliable to unreliable therefore falls in the 8%–16% interval, with 12% marginal and driven by outgoing two-paths.

**Table 6**

*Bias-to-Band Ratio by Effect and Multi-Receiver Proportion ($n = 28$; 12% Cell $n = 26$)*

| MR (%) | Inertia | Reciprocity | OTP | Difference (active) | Same (group) |
|---:|---|---|---|---|---|
| 0 | 0.21 | 0.08 | 0.24 | 0.02 | 0.09 |
| 2 | 0.10 | 0.07 | 0.05 | 0.06 | 0.26 |
| 8 | 0.09 | 0.33 | 0.62 | 0.24 | 0.62 |
| 12 | 0.22 | 0.58 | 0.95 | 0.30 | 0.80 |
| 16 | 0.50 | 0.94 | **1.29** | 0.29 | **1.09** |
| 24 | 0.72 | **1.23** | **1.92** | 0.45 | **1.44** |

*Note.* Each entry is the absolute dataset-averaged bias divided by the per-effect band of Table 5. Values in bold exceed 1 and are out of band. The 12% row is the off-grid descriptive cell and does not enter the threshold calculation; it is based on $n = 26$ rather than $n = 28$ because its baselines were generated in a later batch under a newer `remulate` build, under which the baseline-quality guard of Section 2.6 drops two further low-gap seeds.

## 3.4 The RHEM contextual benchmark

The native RHEM fitted to the same multi-receiver data is reported alongside the dyadic REM, on the raw `eventnet` coefficient scale (Appendix A). Where the dyadic REM attenuates, the RHEM estimate stays close to its baseline value as the multi-receiver proportion grows. Same-group homophily is the clearest case, holding near 0.42–0.43 across the whole grid under RHEM while the dyadic REM falls from $0.39$ at 0% to $0.28$ at 24%; the active-attribute difference and reciprocity show the same pattern more mildly.

Two effects depart from their generating values under RHEM. Outgoing two-paths sits near zero on the raw scale, roughly $0.029$ below the generating value at every multi-receiver proportion; the gap is near-constant across the grid ($-0.029$ at 0% to $-0.030$ at 24%), the signature of a deterministic counting-convention difference between the dyadic and hyperedge implementations rather than a transformation-induced effect (Appendix A). Under RHEM, inertia sits above its generating value (around $0.034–0.038$ against $0.02$); unlike the outgoing-two-paths offset this gap is smaller and drifts mildly across the grid, so we report it as where the coefficient lands on RHEM's raw scale rather than attributing it to a single constant offset. Neither enters the threshold, which is judged against truth on the REM side (Section 2.5); the RHEM column is a contextual reference only.

The contrast is nonetheless informative for the practical use of RHEM. Once the structural coefficients of the dyadic REM begin to attenuate with the multi-receiver share, the native RHEM continues to hold those same coefficients close to their baseline values, which is the behaviour expected of the more faithful model in exactly the range where the dyadic transformation can no longer be relied upon. This is consistent with recommending a native multi-receiver model above the threshold, as set out in the Conclusion and Discussion (Section 4.2).

## 3.5 The baseline rate

The log baseline rate $\beta_0$ is excluded from the acceptability rule (Section 2.7.1); its trajectory is reported here as a check on the $\log(1+p)$ prediction of Appendix B. At the 0% condition $\beta_0$ is recovered close to its generating value ($-5.02$ against $-5.00$). As the multi-receiver proportion increases, the recovered $\beta_0$ rises monotonically, by $0.106$ at 8% and $0.276$ at 24% relative to the 0% mean. The first-order prediction $\log(1+p)$ gives $0.077$ and $0.215$ at the same proportions, so the empirical shift has the predicted sign and shape but runs consistently above it, by about $0.06$ at 24%. This over-shift is the expected consequence of the attenuation reported in Section 3.3: as the positively-signed effects shrink towards zero, their average contribution to the linear predictor falls, so $\beta_0$ has to rise beyond $\log(1+p)$ to keep the fitted rate matched to the $(1+p)\,M$ observed rows. The same-group attenuation alone, a drop of roughly $0.11$ on an indicator with mean near $0.5$, accounts for about $0.05$ of the $0.06$ over-shift at 24%. Folding the coefficient attenuation into the accounting argument of Appendix B therefore reproduces the trajectory, and the exclusion of $\beta_0$ from the threshold rule stands.

## 3.6 Sensitivity to the acceptability rule

The headline threshold is reported under the per-effect recovery-noise band. Re-applying the earlier specification, a single absolute log-rate cap of $|\text{bias}| \leq 0.05$ across all effects (Section 2.7.3), returns the same grid threshold of 8%: at 8% the largest absolute bias is same-group homophily at $0.050$, at the cap edge, and at 12% same-group homophily reaches $0.064$ and fails the cap. The two rules therefore agree on the reliable ceiling at 8%

and agree that deterioration begins in the 8%–16% interval, with the cap rule failing at 12% where the band rule is marginal. The 8% threshold is robust to this change of rule.

# 4. Conclusion and Discussion

The introduction set out a problem that applied researchers face whenever they work with communication data: the relational event model is dyadic, much real communication is multi-receiver, and the common workaround of replicating each multi-receiver event into several dyadic events is applied routinely but without a quantitative sense of when it stops being trustworthy. This thesis set out to close that gap by measuring, in a controlled simulation with known generating parameters, the multi-receiver share up to which the dyadic transformation still recovers those parameters. This chapter interprets what that measurement means, where it applies, and where it does not. It does not introduce new results; the numbers it refers to are reported in the Results chapter.

## 4.1 The multi-receiver threshold for parameter recovery

The core finding is a threshold. Measured against the per-effect recovery-noise band over 28 independent baseline datasets, the dyadic transformation recovers the known generating parameters up to a multi-receiver share of about 8%, becomes borderline through roughly 12%, and is no longer reliable beyond it (Section 3.2, Section 3.3). What makes the finding more than a single number is the order in which the effects fail, which follows directly from how the transformation contaminates each statistic.

Outgoing two-paths is the first to leave its band, and it does so because it is triadic. Each injected sender-to-extra-receiver event creates a spurious two-path for every past event into that receiver, so triadic contamination accumulates faster than the contamination of any dyadic statistic and outruns its own very tight recovery noise first, despite only a small absolute bias. Same-group homophily attenuates next because the extra receiver is drawn at random and is group-agnostic, so it dilutes the homophily signal towards zero. Reciprocity picks up spurious returns whenever the extra receiver had previously messaged the sender, while inertia and the active-attribute difference are the most local to the original dyad and dilute slowest. The threshold is therefore not an arbitrary cut-off but the point at which the most contamination-prone statistic, the triadic one, can no longer be distinguished from its own recovery noise.

## 4.2 A three-zone operating range and the recommendation for applied work

Section 1.5 asked up to what proportion of multi-receiver events the dyadic transformation preserves REM estimates close to the known generating parameters. The answer is best expressed as a three-zone operating range rather than a single cut-off. The transformation is reliable up to a multi-receiver share of about 8%, where every effect is comfortably within its band; borderline between 8% and roughly 12%, where the rule still passes but the binding statistic, outgoing two-paths, is within band-uncertainty of failing; and unreliable above roughly 12%, where outgoing two-paths crosses its band, with the breach unambiguous by 16% when same-group homophily moves clearly out of band as well.

The recommendation for applied work follows directly. For multi-receiver shares up to about 8% the dyadic transformation with a tie-breaking timestamp noise recovers the generating parameters well enough to be relied upon; in the 8%–12% range it should be used with caution, with the multi-receiver share reported alongside the estimates; and above roughly 12% a native multi-receiver model such as RHEM (Lerner & Lomi, 2023), a restructuring of the data, or a different estimator should be used instead of the dyadic transformation. The contextual RHEM benchmark supports this directly: in exactly the range where the dyadic REM begins to attenuate, the native model holds its structural coefficients close to their baseline values (Section 3.4).

## 4.3 Implications for the applied domain

Within the range studied, the recommendation is a guardrail rather than a repair: above the borderline zone the dyadic transformation should be replaced, not patched, because the receiver-set composition lost in separating an event into dyads is not recoverable from the transformed data. The recommendation is also bounded above by the 24% grid, whose upper end already sits below the 31% multicast share that Mulder and Hoff (2024) report for the Enron corpus; the experiment makes no claim about higher shares.

The behaviour of the baseline rate has a direct applied reading. Additive injection makes the long table genuinely larger, so the recovered pacing constant rises with it (Section 3.5); this is the in-vitro counterpart of the sample-size inflation that Mulder and Hoff (2024) identify when carbon-copy lists are dyadically expanded. An analyst who dyadically expands real multicast data and reports the recovered baseline rate is therefore describing the expanded process, not the underlying multicast process, and should read it as such. The thesis recommendation concerns recovery of the tie-oriented coefficients, which carry the substantive content of an applied REM analysis.

How far the 8% ceiling carries to real multicast data depends on how real multicast differs from the random-tagging design used here. The extra receivers in this design are drawn uniformly from non-participating actors and carry no covariate or history signal, which is arguably the most benign multicast structure available: it adds rows without introducing receiver-set composition that the dyadic REM cannot represent. On the a-fortiori reading, structured multicast can then only fare the same or worse, so a ceiling of about 8% even under maximally uninformative multicast is a defensible upper bound on where recovery remains safe. The competing reading is that structured multicast biases the dyadic statistics differently rather than simply more: correlated carbon-copy lists could, for instance, attenuate same-group homophily less if the co-receivers share the sender's group, while fat-tailed receiver-set sizes would multiply exactly the spurious two-paths that already make outgoing two-paths the binding statistic. Because the binding statistic is triadic, and correlated receiver sets would plausibly create more spurious two-paths rather than fewer, we treat the a-fortiori reading as the load-bearing one for a conservative recommendation, while noting that a structured-multicast generator is the natural experiment to settle the direction.

## 4.4 Limitations

Several scope conditions bound the result. The simulation is run at a single configuration ($N = 20$ actors, $M = 2{,}000$ events), so the threshold is conditional on this configuration. The dependence on $M$ is not harmless: cumulative-count inertia is self-reinforcing, so the feasible inertia coefficient shrinks as $M$ grows (Table 3), and an exploratory check at $M = 3{,}000$ made recovery worse across every effect through the same feedback. A larger event count is therefore not a route to a safer transformation, and the 8% ceiling should be re-derived for a substantially different ($N, M$,inertia) setting rather than transported as a constant. Every multi-receiver event here has exactly two receivers, so receiver-set cardinality is not varied. The small generating value of inertia is a property of cumulative-count inertia specifically; a decaying-memory operationalisation would bound the statistic and tolerate larger coefficients, and is not exercised here. Finally, the simulation assumes constant generating parameters, whereas coefficients in real relational-event processes can drift over time, so the recommendation should be read as a statement about stationary parameter recovery rather than about tracking time-varying effects.

## 4.5 Future work

Each limitation is a natural axis for future work. The most immediately informative is a structured-multicast generator, which would replace the uninformative random tagging with correlated carbon-copy lists and fat-tailed receiver-set sizes and so settle whether structured multicast shifts the threshold upward or downward. Beyond that, re-deriving the threshold across a grid of $(N, M)$ configurations would establish how it scales with network size and history length; varying receiver-set cardinality would extend the result beyond two-receiver events; and introducing time-varying generating parameters would connect the result to the dynamic REM extensions used in applied work.

## 4.6 Ethical implications and considerations

The study itself raises no data-subject concerns: it uses only synthetic data, with no human participants, no personally identifiable information, and no restricted third-party data. The ethical weight of the work is in how its recommendation is used. Dyadic transformation inflates the apparent sample size, which deflates standard errors and can produce overconfident inferences (Mulder & Hoff, 2024); an analyst who transforms heavily multi-receiver data and reads the result at face value risks reporting effects, and supporting decisions that rest on them, with more certainty than the data warrant. The threshold is a guardrail against this, and using it responsibly means reporting the multi-receiver share of a dataset and justifying the modelling choice against it rather than transforming silently. Because the evidence here comes from a single synthetic setting, the specific 8% figure should be treated as a calibrated guideline rather than a universal constant, and applied with the scope conditions of Section 4.4in view.

## 4.7 Conclusion

This thesis began with a gap that applied researchers routinely walk into without noticing: the dyadic transformation of multi-receiver communication data is convenient and widely used, yet there has been no quantitative sense of when it stops recovering the model that generated the data. By simulating from known parameters and judging recovery against a pre-specified per-effect band, the thesis turns that open question into a concrete operating range. Dyadic transformation with a tie-breaking timestamp noise is reliable up to a multi-receiver share of about 8%, becomes borderline up to roughly 12%, and should be replaced by a native multi-receiver model above it, with the triadic outgoing-two-paths effect the first to fail. The figure is a calibrated guideline rather than a universal constant, but the mechanism behind it, and the recommendation to report the multi-receiver share and choose the model accordingly, are general. The result gives applied researchers a defensible reason to either keep using the familiar dyadic toolchain or to step beyond it, and a clear marker for which side of that line a given dataset falls on.

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## Appendix A. RHEM estimation configuration

This appendix records the `eventnet` and `survival` configuration that produces the RHEM coefficient estimates summarised at a higher level in Section 2.5 of the Methodology chapter.

The pipeline has two stages. In the first stage, the `eventnet` software (Lerner & Lomi, 2023) enumerates the hyperedge risk set from the multi-receiver long table. For each observed hyperevent and a case-control sample of non-events, `eventnet` emits the hyperedge covariates that correspond to the five dyadic statistics fitted on the REM side: inertia, reciprocity, outgoing two-paths, dyadic attribute difference on the continuous attribute, and same-group homophily on the binary attribute. The mapping between the dyadic `remstats` definitions and the corresponding `eventnet` effect families is the one referenced in Section 1.4 of the Introduction.

In the second stage, the model is fitted by stratified Cox regression in the `survival` package (Therneau, 2023). The stratification convention is the one used in the official multicast-RHEM tutorial of Lerner and Lomi: the `eventnet` output assigns one stratum per hyperevent through the `EVENT_INTERVAL` variable, and coxph is called with `strata(EVENT_INTERVAL)` so that each case-control set of one observed hyperevent and its sampled non-events is matched within its own stratum, the convention that places the estimates on the RHEM coefficient scale of Lerner and Lomi (2020, 2023).

One scale caveat applies to the outgoing-two-paths effect. The dyadic `remstats` outgoing-two-paths count and the hyperedge outgoing-two-paths count produced by `eventnet` are not on an exactly aligned raw scale: the two implementations differ in how they aggregate over intermediate actors when constructing the path count. A constant offset across MR levels would indicate a deterministic implementation difference, whereas variation with $p$ would indicate a transformation-induced effect.

## Appendix B. Derivation of the $\log(1+p)$ shift in $\beta_0$

This appendix gives the derivation summarised in Section 2.7.1 of the Methodology chapter.

The additive multi-receiver construction of Section 2.2 leaves every baseline event in place and adds one extra row per tagged event, so at multi-receiver proportion $p$ the long-table row count grows from $M$ at $\mathrm{MR} = 0$ to $M(1+p)$. Over a fixed observation window the apparent per-dyad rate inflates by a factor of $(1+p)$, and to first order the recovered log baseline shifts by $\log(1+p)$. Numerically, this predicts a shift of $\log(1.02) \approx 0.020$ at $p = 2\%$, rising to $\log(1.24) \approx 0.215$ at $p = 24\%$.

This first-order prediction is exact in the null model (every tie-oriented statistic identically zero), where there is no structural contribution to redistribute. Away from the null it is an approximation, because the injected events are not structurally inert. The extra receivers of the uninformative-receiver design of Section 2.2 carry no covariate or history signal of their own, so they introduce no genuine new effect to estimate; but the act of injecting their events still mechanically contaminates the endogenous statistics, adding spurious two-paths and reciprocation and diluting the homophily signal (Section 4.1). The tie-oriented coefficients therefore attenuate towards zero (Section 3.3), and as their positive average contribution to the linear predictor shrinks, $\beta_0$ rises slightly beyond $\log(1+p)$ to keep the fitted rate matched to the observed events (Section 3.5). The dominant share of the rate inflation is still absorbed by $\beta_0$, and the residual over-shift is a fingerprint of that attenuation rather than error in the $\log(1+p)$ prediction. Under structured multicast the redistribution would be larger and could load onto particular tie-oriented coefficients (for example inertia, if the same group of receivers is addressed together repeatedly).

The shift is therefore an accounting consequence of additive injection, not a recovery failure of the dyadic REM under multi-receiver structure: $\beta_0$ is responding to a long table that genuinely contains $(1+p)M$ events, and it recovers the pacing constant of that augmented process. Including $\beta_0$ in the rule would answer a different question (how much does the dyadic event count inflate per extra receiver?) than the one this thesis poses (when does the dyadic REM stop recovering the multi-receiver-perturbed generating process?). The empirical $\beta_0$ trajectory is reported in the Results chapter as a check on the $\log(1+p)$ prediction and on the uninformative-receiver assumption, but it does not enter the threshold.

## Appendix C. Code and data availability

All code used in this thesis is publicly available in the project repository at https://github.com/GabrielSC92/Thesis-MSc-ADS-4266390. The repository contains the synthetic data-generation scripts (the baseline data-generating process and the multi-receiver construction), the tie-breaking transformation, the dyadic REM and RHEM estimation pipelines, the cross-dataset replication harness, and the analysis notebooks that derive the per-effect bands, apply the acceptability rule, and produce every table and figure reported here. Because the full pipeline is deterministic in the baseline seed (Section 2.6), every number in the Results chapter can be regenerated from the seed list alone.

# Appendix D. Model AI Disclosure Statement

*Version January 2026*

I, Gabriel Silva Cheinquer, 4266390, hereby confirm

that the present work is my original work. I declare that I have used AI tools only as allowed for this work.

Please indicate if you used any AI tools in your assignment:

- (●) Yes, I used AI tools to assist me with the present work.
- ( ) No, I did not use any AI tools to assist me with the present work.

If you selected "Yes", please indicate the following.

Which AI tool(s) did you use and for what purpose?

| Name of the AI tool | Purpose of the AI tool deployment (e.g. planning, brainstorming, editing, outlining, feedback, content generation of <specify>, debugging code etc) |
|---|---|
| Cursor, Claude Opus 4.6-4.8 | Planning and project tracking (organising my own decision logs and draft versions), feedback and editing, content generation of code + debugging. |
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| | |

I acknowledge my responsibility as a student/learner to thoroughly verify all outputs and content produced by AI tools and accept full accountability for their accuracy and validity.

Any comments:

Signature of the student:

Gabriel Silva Cheinquer

Date: 25-06-2026